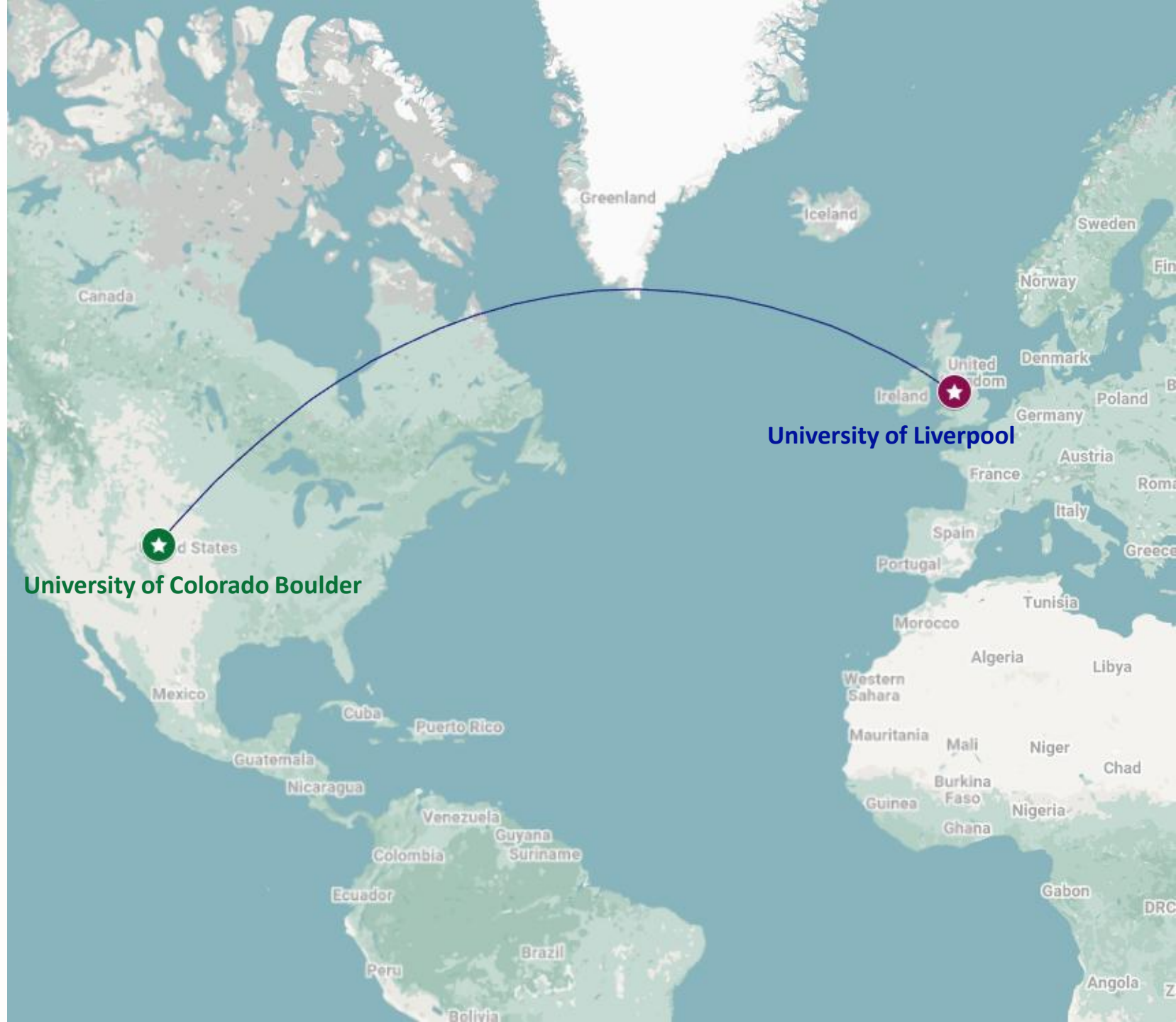


# Asymmetrically Discounted Stochastic Games



**Sarvin Bahmani**  
University of Liverpool

Soumyajit Paul, Sven Schewe,  
Shadi Kalat, Ashutosh Trivedi.

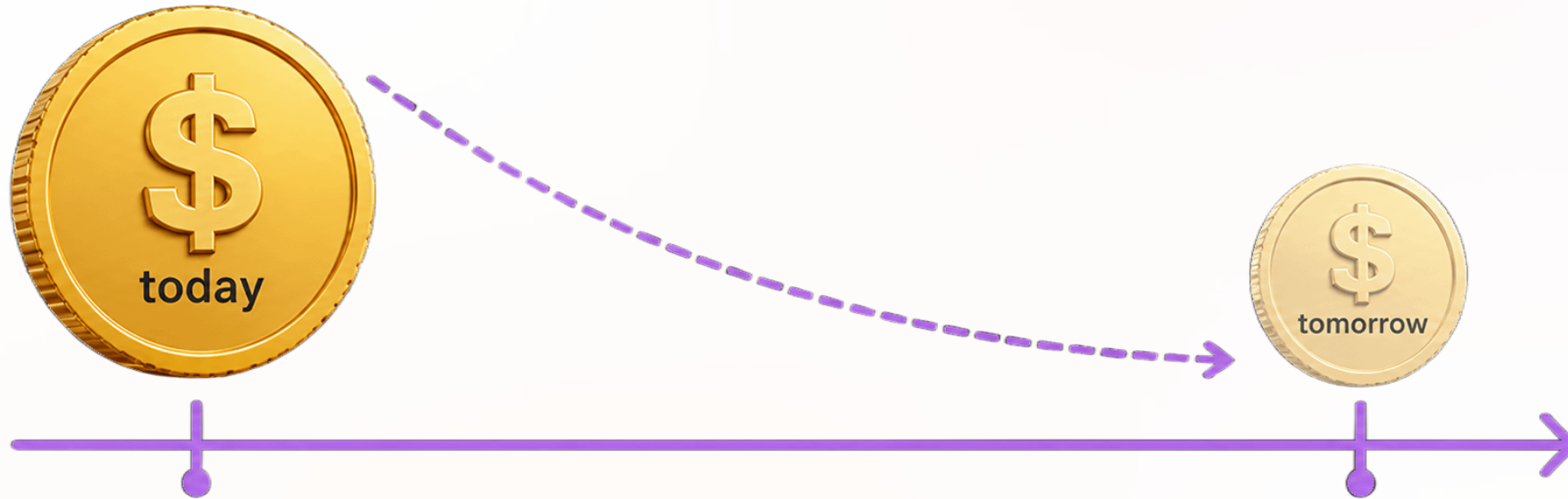


# Two Player Games

How do the contracting parties  
evaluate the rewards  
in the long run?



# The Value of Delayed Rewards



A reward becomes **less valuable** when it is delayed.

# Do people value the future differently?



## Classical Setting

Different player have the  
**same** discount factor.

## Our Setting

Different players discount future  
rewards **differently**.



# How can a bank benefit from this difference?

Nash equilibria?

Stackelberg equilibria?



# Two Player Games

**Maximiser**

Leader

Bank

**Minimiser**

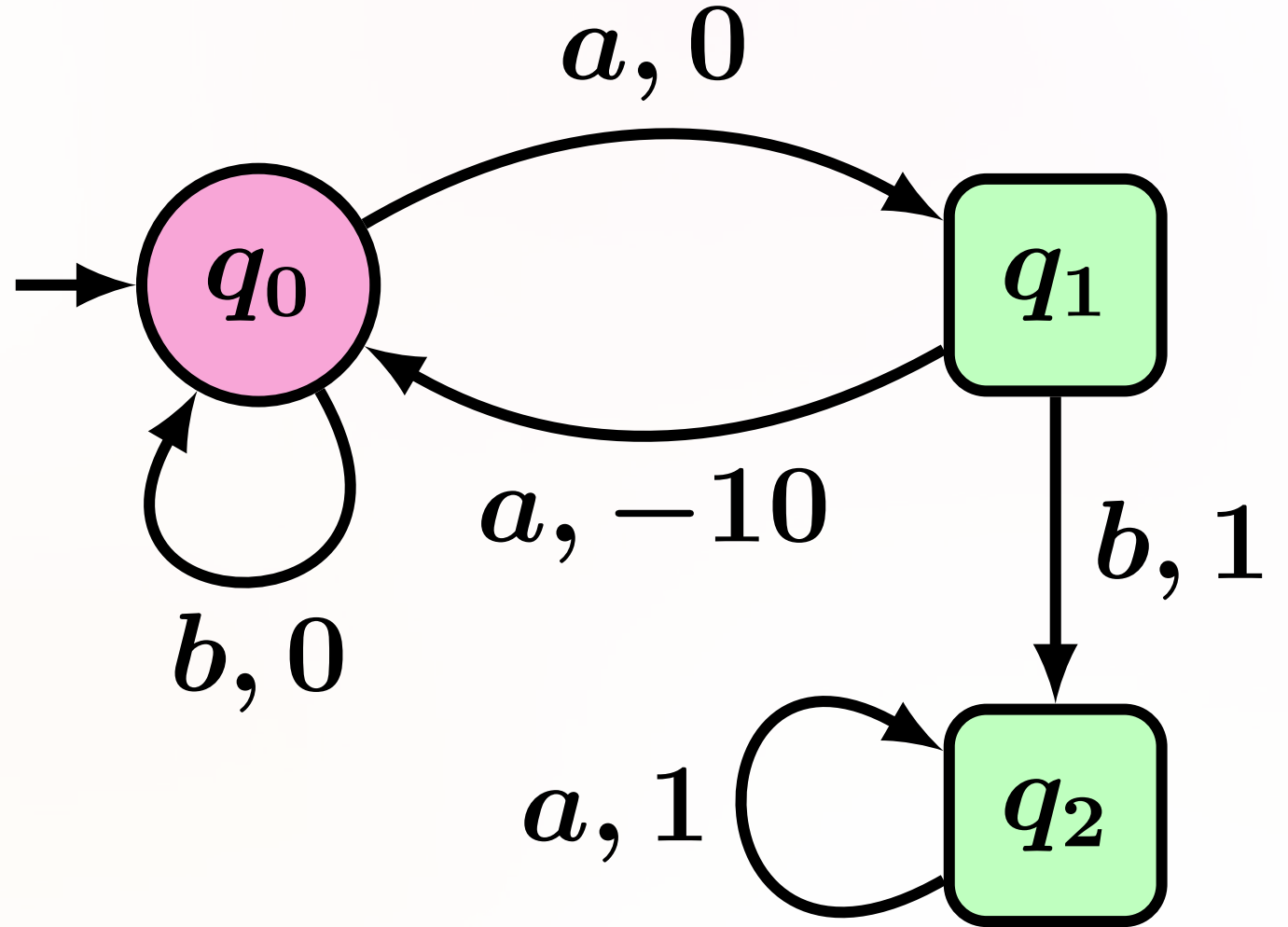
Follower

Customer

# Will customer open an account?

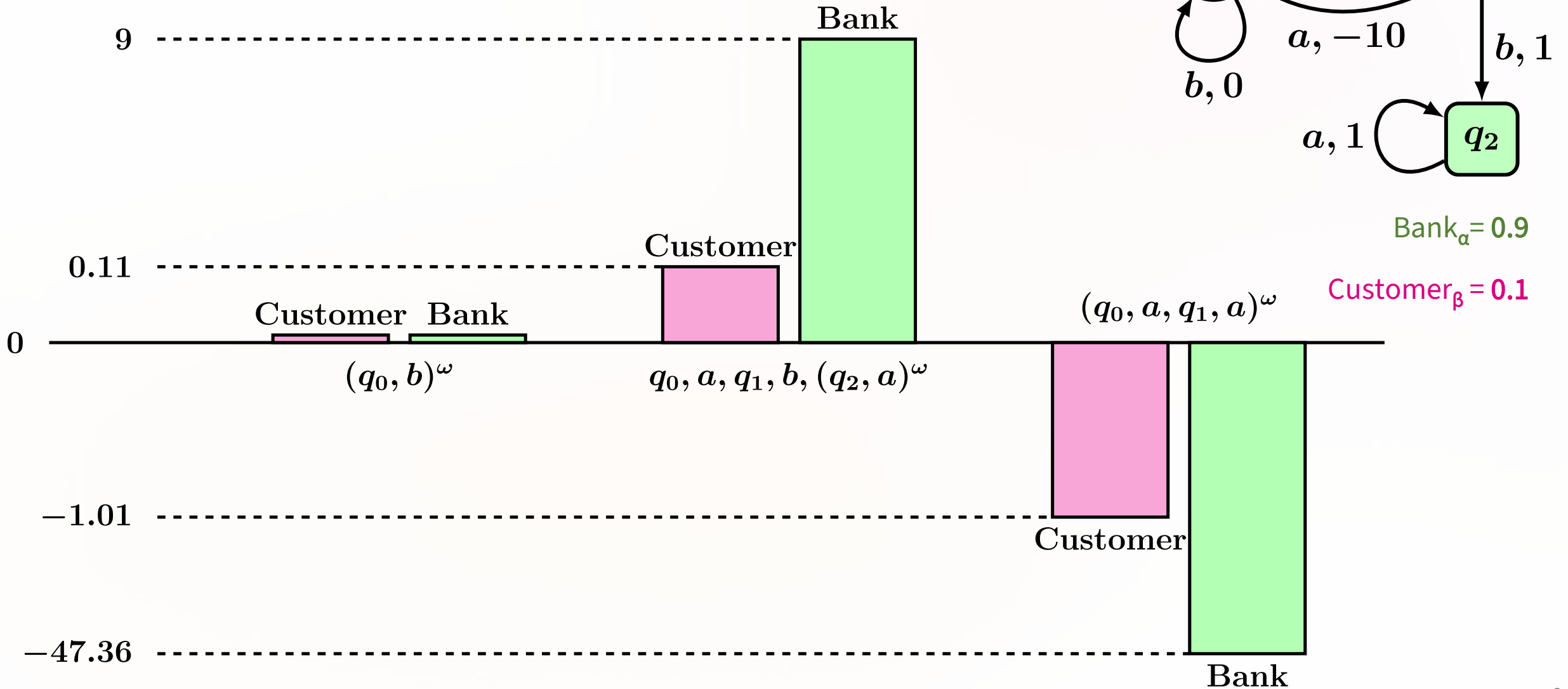
Max/Bank <sub>$\alpha$</sub>  = 0.9

Min/Customer <sub>$\beta$</sub>  = 0.1





# Different Scenarios



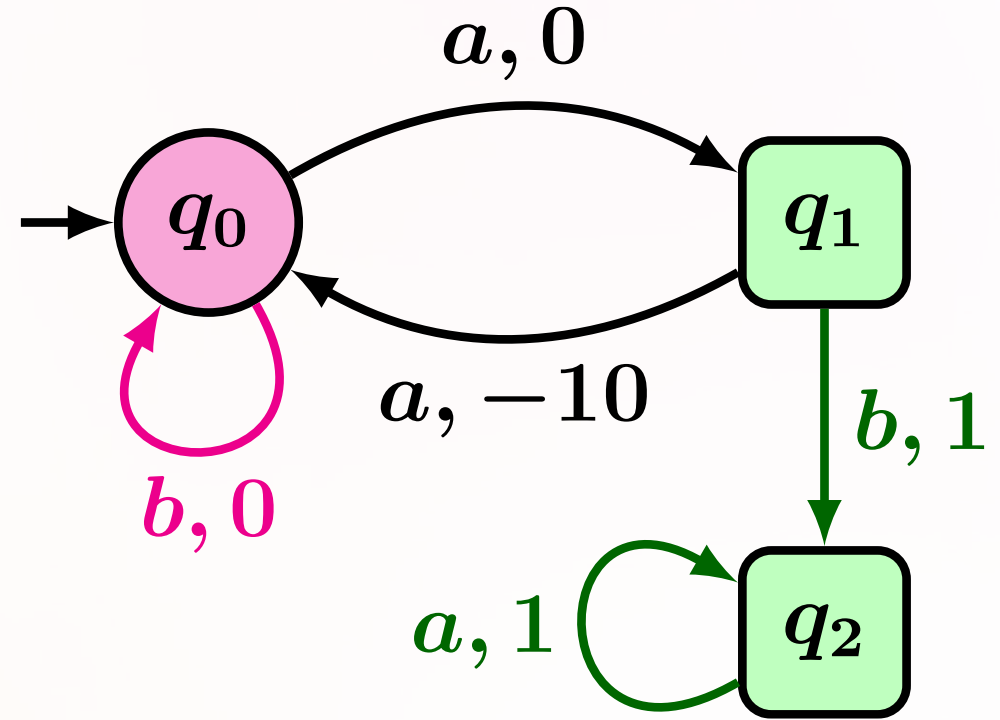
# Customer is not taking the bait.

Bank optimal action at  $q_1$ :  $b, 1$

Customer optimal action at  $q_0$ :  $b, 0$

Bank payoff = 0

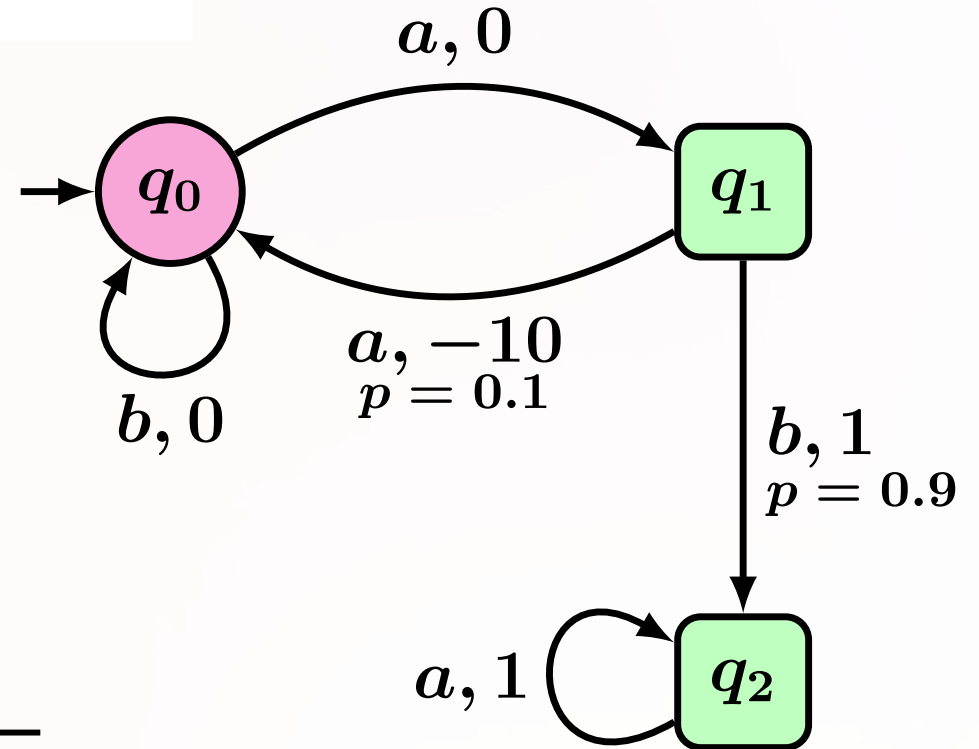
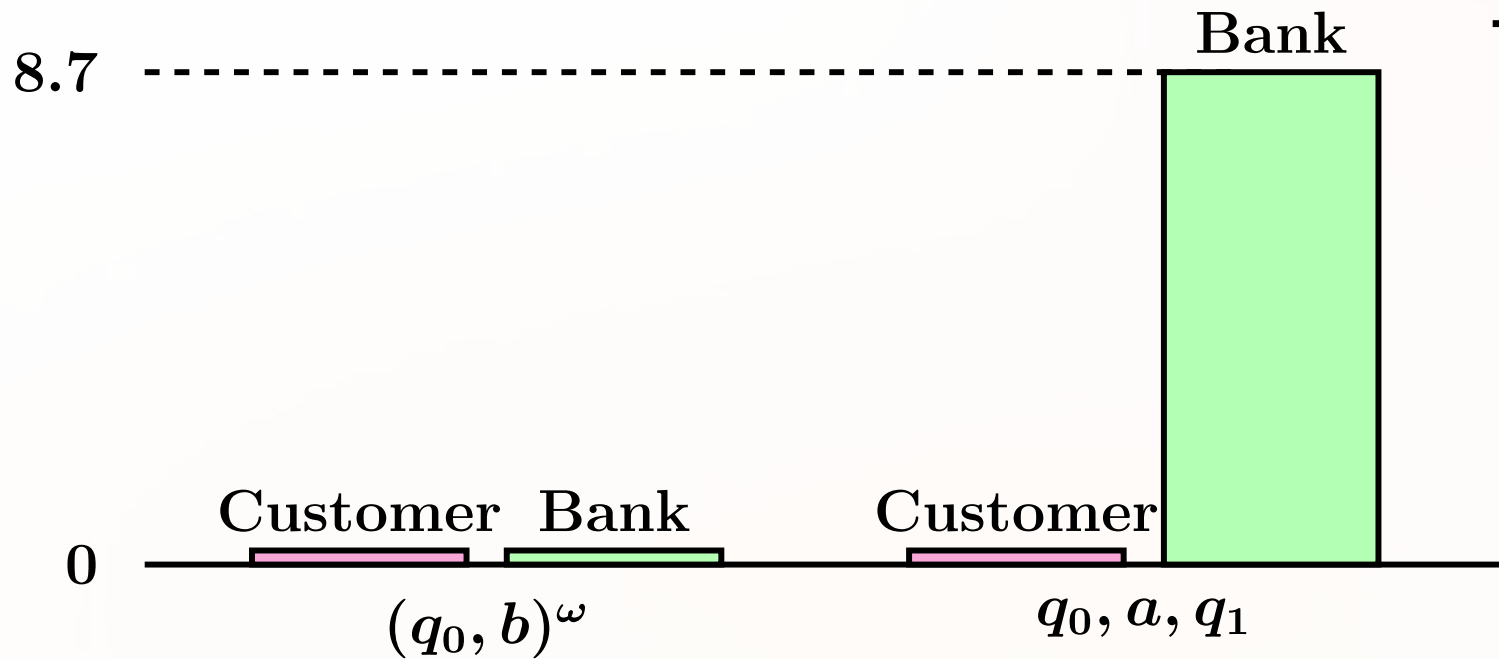
Customer payoff = 0



Bank <sub>$\alpha$</sub>  = 0.9

Customer <sub>$\beta$</sub>  = 0.1

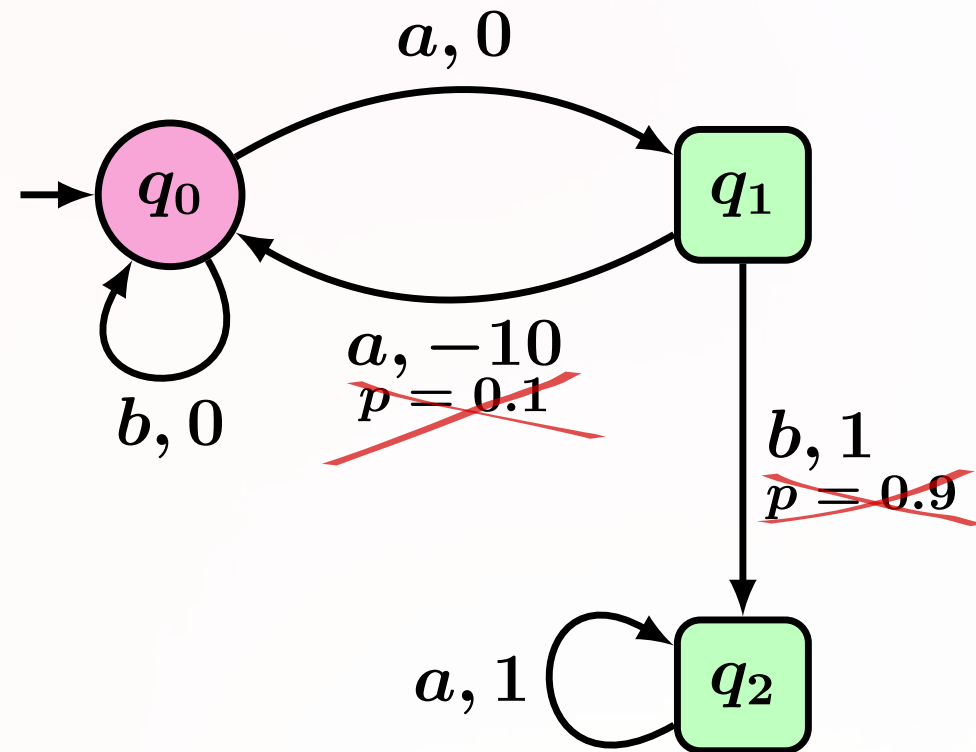
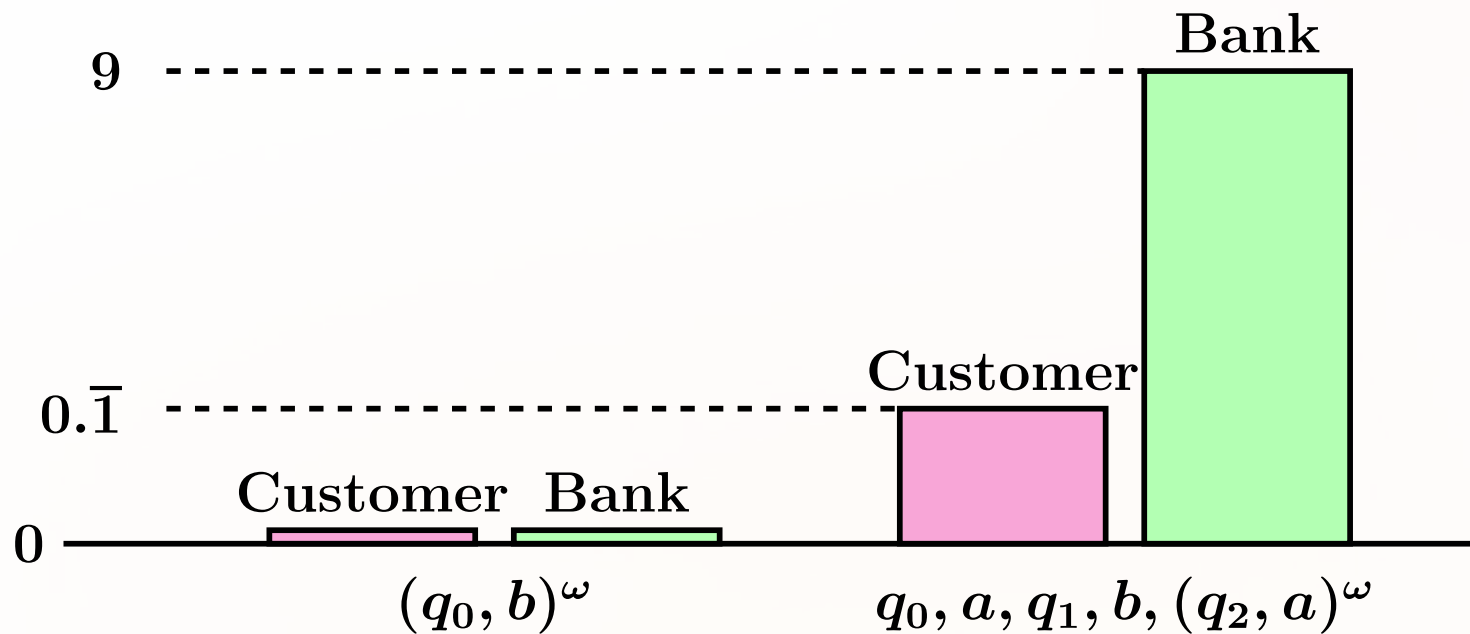
# Randomisation can help. Lottery!



Bank <sub>$\alpha$</sub>  = 0.9

Customer <sub>$\beta$</sub>  = 0.1

no Lottery, *but...*



$\text{Bank}_\alpha = 0.9$

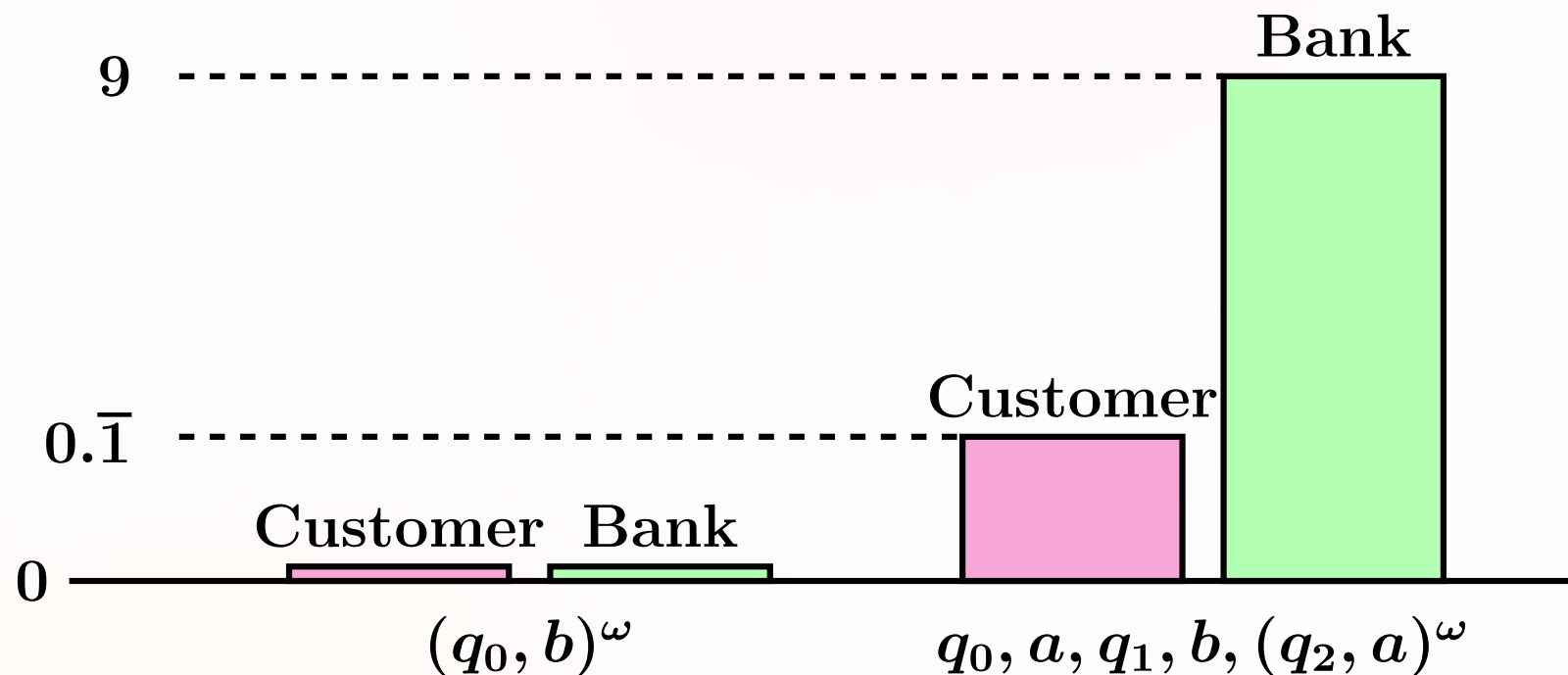
$\text{Customer}_\beta = 0.1$

# no Lottery, *but* Interest!

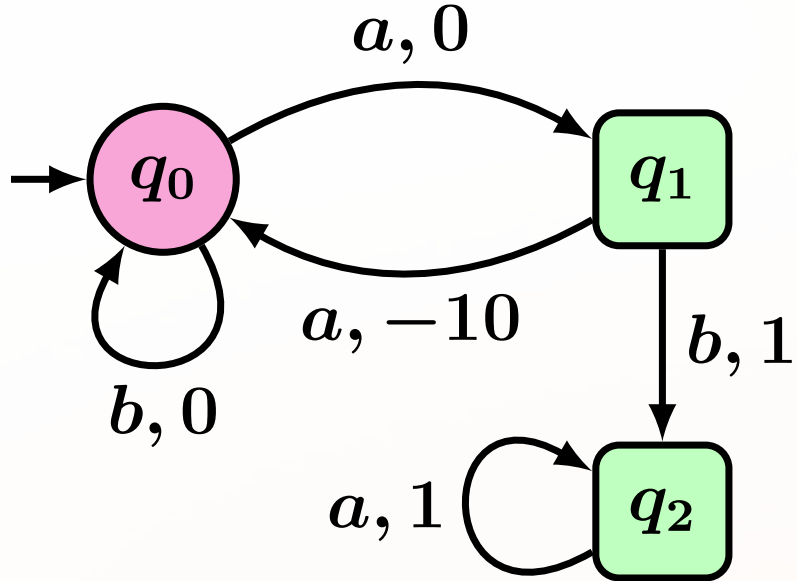
customer Payoff = 0

interest =  $0.\bar{1}$

bank Payoff =  $9 - 0.\bar{1} = 8.\bar{8}$



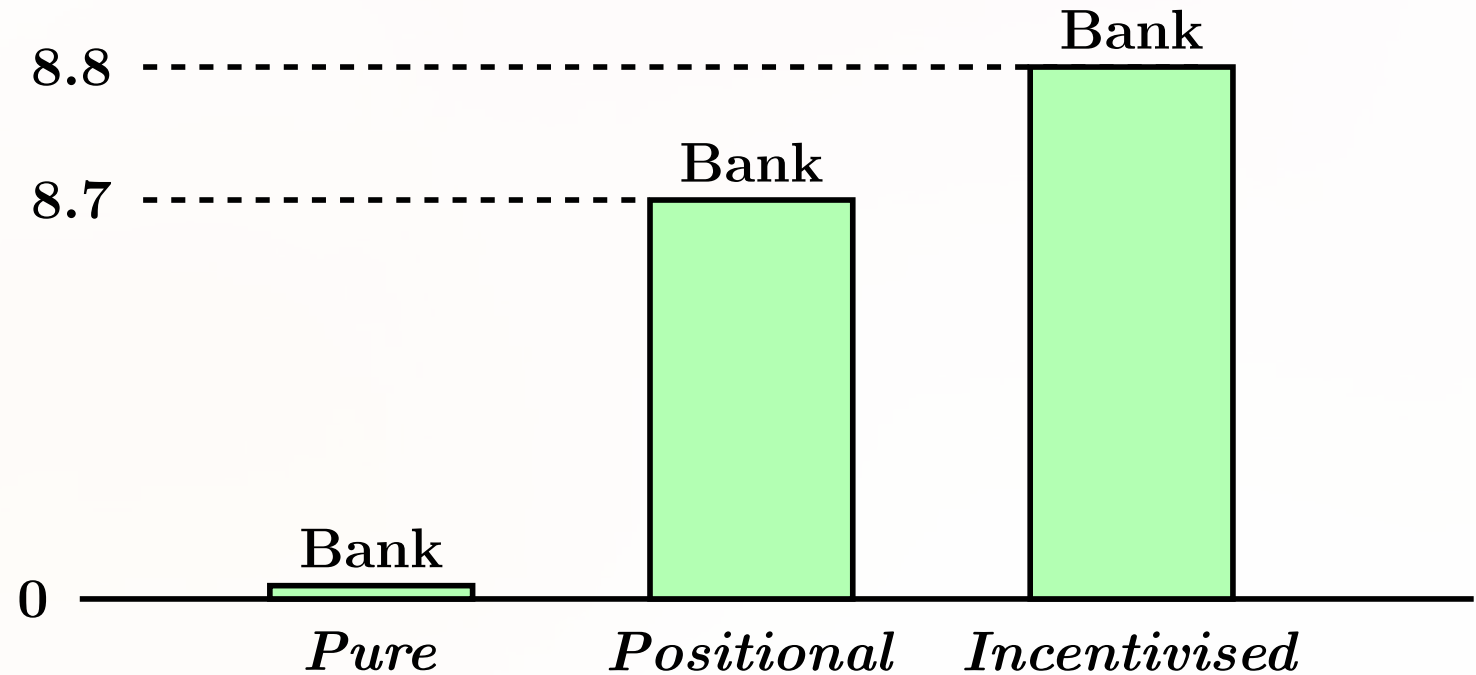
# Bank payoff in different strategies



$\text{Bank}_\alpha = 0.9$

$\text{Customer}_\beta = 0.1$

customer Payoff = 0





# How can a bank benefit from this difference?

~~Nash equilibria.~~

Stackelberg equilibria

Incentive Stackelberg equilibria



# Strategy that bank pays the least interest while customer doesn't lose.

## Negative Results

- Pure stationary strategies?
- Randomised strategies?

Simple stationary behavior is NP-hard

## Positive Results

- ✓ Optimal strategies exist in a simple class.
- ✓ Finite-memory counting strategies

Structured finite memory is  $UP \cap coUP$ .

# Long-term Behaviour: Max Optimal Stationary Tail

We have:  $\alpha > \beta$

0. Find Min payoff

1. Find Max optimal policy

Find Max value and Min value

2. Incentive = Min value - Min payoff

3. Max payoff = Max value - Incentive

# Deviation from Max Stationary Tail

$$\text{MaxDisAdv} = R(s, a) + \alpha \sum_{s'} p(s' | s, a) V_{\text{MAX}}(s') - V_{\text{MAX}}(s)$$

After  $t$  steps

$$\text{MaxDisAdv}_t(s, a) = \alpha^t \text{MaxDisAdv}(s, a)$$

$$\text{MinAdv} = R(s, a) + \beta \sum_{s'} p(s' | s, a) V_{\text{MIN}}(s') - V_{\text{MIN}}(s)$$

After  $t$  steps

$$\text{MinAdv}_t(s, a) = \beta^t \text{MinAdv}(s, a)$$

$$\text{Benefit}_t(s, a) = \beta^t \text{MinAdv}(s, a) - \alpha^t \text{MaxDisAdv}(s, a)$$

# After a finite horizon, deviations cannot help

for every deviation, there exists a finite horizon  $k$  such that, for all  $t \geq k$ :

$$\mathbf{Benefit}_t(\mathbf{s}, \mathbf{a}) \leq 0$$

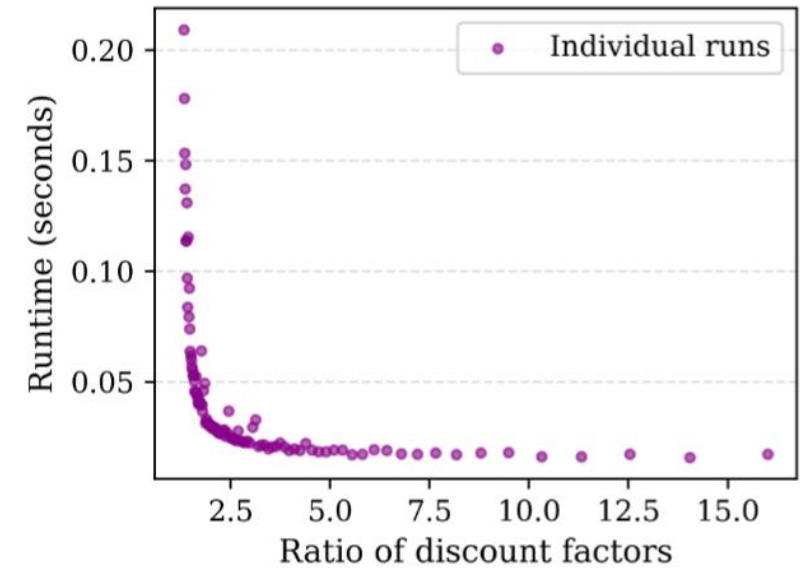
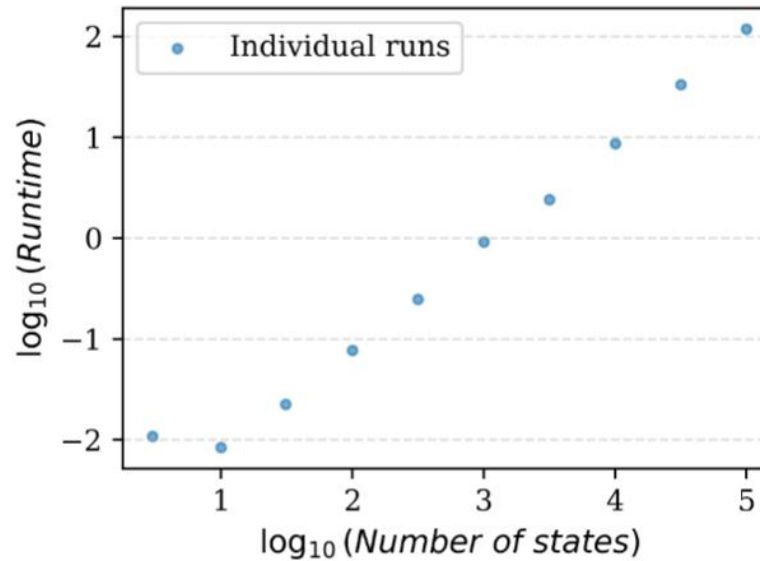
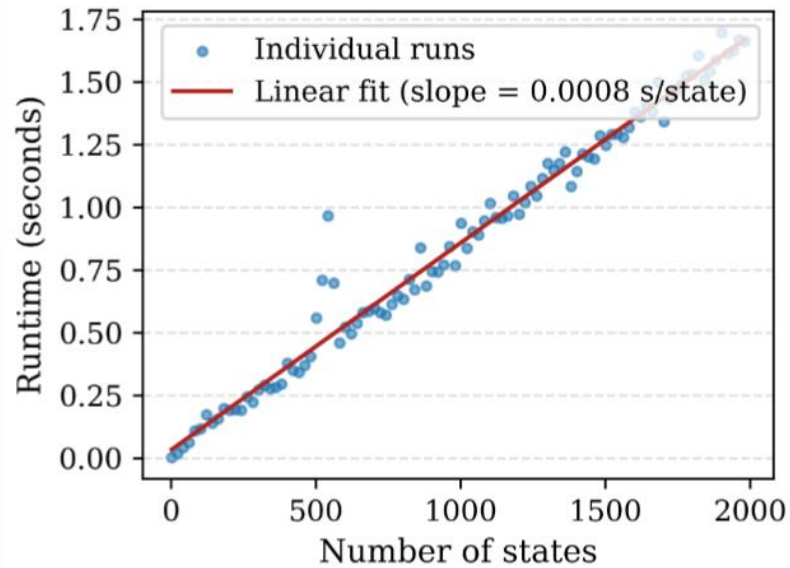
## ❗ **Why $k$ is small under spacing?**

A typical bound has the shape  $k = \mathcal{O}\left(\frac{\log(\text{value ratio})}{\log(\alpha/\beta)}\right)$

## ✅ **Outcome**

Only the prefix before  $k$  needs time-dependent optimisation; after that, follow the stationary tail.

# Scalability



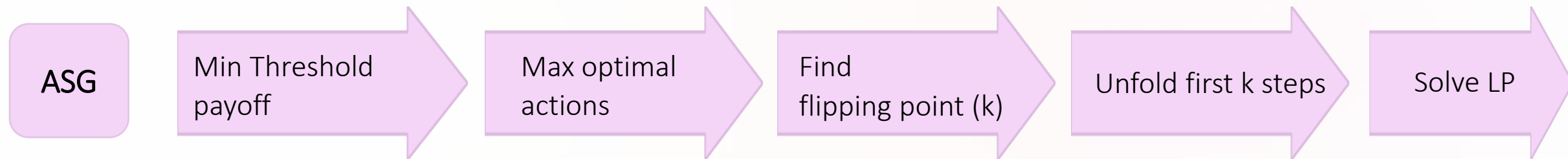
closer discount factor

larger  $\kappa$

larger LP

slower runtime

# Synthesis Algorithm



## Deviation benefit:

$$\text{Benefit}_k(s, a) = \beta^k \text{MinAdv}(s, a) - \alpha^k \text{MaxDisAdv}(s, a)$$

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$$\pi^* = \underbrace{\text{[finite prefix]}}_{\text{counting memory}} \cdot \underbrace{\left[ \pi_{\text{long}}^\omega \right]}_{\substack{\text{stationary tail} \\ \text{(Max policy)}}}$$

# Complexity Contrast

| Stationary strategies                                                                                                         | Finite counting memory                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Too restrictive:</p> <ul style="list-style-type: none"><li>▪ Loses Optimal Policy</li><li>▪ Synthesis is NP-hard</li></ul> | <p>Sufficient for incentivised Stackelberg:</p> <ul style="list-style-type: none"><li>✓ Attains optimal Policy</li><li>✓ Synthesis is <math>UP \cap coUP</math></li></ul> |



# Takeaways

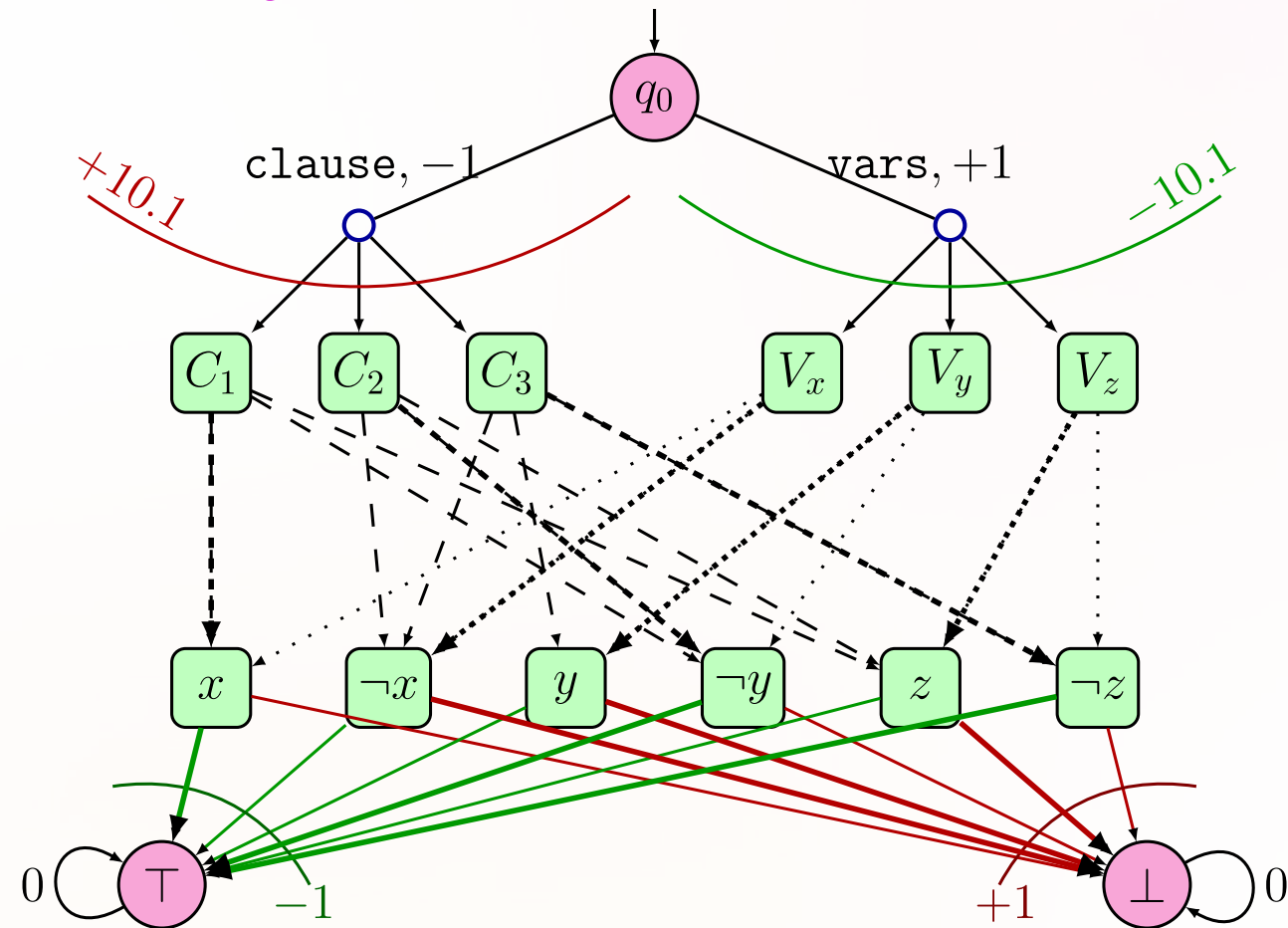
*Thank you!*

- 1 — Asymmetric discount factor change the temporal structure of optimal policy.
- 2 — Optimal policy may need memory.
- 3 — Optimal strategies have a simple form:  $\pi^* = \underbrace{[\text{finite prefix}]}_{\text{counting memory}} \cdot \underbrace{[\pi_{\text{long}}^\omega]}_{\text{stationary tail}}$
- 4 — Under reasonable spacing, synthesis is tractable.



Different notions of patience do not merely reweight rewards;  
they change the structure of optimal behaviour.

## NP-Hardness, Stationary



## NP-Hardness, Incentivised

